RAG Learning

1.1 Proof

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs p1–p3

What this topic covers. Three ways of *establishing* that a statement is true — deduction, exhaustion and contradiction — and one way of establishing that it is *false* — the counter-example. MEI expects you to understand the structure of a mathematical argument, not merely to reproduce two famous proofs.

Not in this specification: proof by *induction*. It belongs to Further Mathematics, not to H640. If a question seems to demand induction you have misread it — look for a deduction or contradiction route instead.

1. The language of proof

Before any proof, you need the vocabulary that carries the logic. Marks are routinely lost for using these terms loosely.

Term	Meaning
Axiom	A statement assumed true without proof.
Conjecture	A statement believed to be true but not yet proved.
Theorem	A statement that has been proved.
Lemma	A smaller proved result used as a step towards a theorem.
Corollary	A result that follows quickly from a theorem just proved.

Counter-example	A single case for which a general statement fails.

Notation

Symbol	Read as	Meaning
$P \Rightarrow Q$	"P implies Q"	If P is true then Q is true.
$P \Leftarrow Q$	"P is implied by Q"	If Q is true then P is true.
$P \Leftrightarrow Q$	"P if and only if Q"	Both implications hold.
$\forall$	"for all"	The statement holds for every case.
$\exists$	"there exists"	At least one case works.

The single most common logical error at A-Level is treating $\Rightarrow$ as though it were $\Leftrightarrow$. From $x = 2$ you may deduce $x^2 = 4$, but from $x^2 = 4$ you may *not* deduce $x = 2$, because x could be -2. Whenever you write $\Leftrightarrow$, check that *both* directions really hold.

Converse and contrapositive

Given the implication "if P then Q":

- The **converse** is "if Q then P". It is a *different* statement and need not be true.
- The **contrapositive** is "if not Q then not P". It is *logically equivalent* to the original — always true exactly when the original is.

Example. "If n is a multiple of 4 then n is even."
Converse: "If n is even then n is a multiple of 4." — **false** (take $n = 6$).
Contrapositive: "If n is odd then n is not a multiple of 4." — **true**, as it must be.

The contrapositive is a genuine proof strategy: if a direct argument is awkward, proving the contrapositive proves the original. It is often the neatest route to statements of the form "if n^2 is even then n is even".

2. Proof by deduction

Proof by deduction proceeds from given assumptions through a series of logical steps to a conclusion.

This is the workhorse method. The skill is choosing an algebraic representation general enough to cover every case.

Description	Algebraic form
An even number	$2n$
An odd number	$2n+1$
Two consecutive integers	$n,\ n+1$
Three consecutive integers	$n-1,\ n,\ n+1$
A multiple of k	kn
Two *different* even numbers	$2a,\ 2b$ — **not** $2n,\ 2n$

Worked example. Prove that the sum of any two consecutive integers is odd.
Let the integers be n and $n+1$, where n is an integer.

$$n+(n+1)=2n+1$$

Since n is an integer, $2n$ is even, so $2n+1$ is odd. Therefore the sum of any two consecutive integers is odd. ■

Worked example. Prove that the sum of three consecutive integers is always divisible by 3.
Let the integers be $n-1$, n and $n+1$.

$$(n-1)+n+(n+1)=3n$$

$3n$ is a multiple of 3 for every integer n. ■

Choosing the middle integer as n makes the algebra collapse immediately; starting at n and writing $n + (n + 1) + (n + 2) = 3n + 3$ works too, but needs the extra line $3n + 3 = 3(n + 1)$.

Two habits that earn the final mark: (1) **state the generality** — "let n be any integer" — at the start; (2) **state the conclusion in words** at the end. A page of correct algebra with no concluding sentence is regularly capped in MEI mark schemes.

3. Proof by exhaustion

Proof by exhaustion establishes a statement by checking every case of a *finite* set of cases.

The method is only available when the cases can be reduced to a finite list. The art lies in that reduction: usually you split the integers into a small number of classes rather than testing numbers one at a time.

Worked example. Prove that n^2 leaves remainder 0 or 1 on division by 4, for every integer n. Every integer is either even or odd — two cases, which exhausts all possibilities.

Case 1: n even. Write $n = 2k$. Then $n^2 = 4k^2$, a multiple of 4, so the remainder is 0.

Case 2: n odd. Write $n = 2k + 1$. Then

$$n^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1$$

so the remainder is 1.

The two cases cover every integer, so the statement holds. ■

Worked example. Prove that if 3 does not divide n, then n^2 leaves remainder 1 on division by 3. If $3 \nmid n$ then n is of the form $3k + 1$ or $3k + 2$ — two cases.

Case 1: $n = 3k + 1 \Rightarrow n^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1.$

Case 2: $n = 3k + 2 \Rightarrow n^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1.$

In both cases the remainder is 1. ■

Always say explicitly *why your cases are exhaustive* — "every integer is either even or odd", "an integer not divisible by 3 must be of the form $3k+1$ or $3k+2$". Without that sentence the proof has a hole, and examiners look for it.

4. Disproof by counter-example

To **disprove** a general statement it is enough to exhibit a **single** case for which it fails.

Note the asymmetry: one example never proves a general statement, but one counter-example always disproves it. Show the arithmetic — a bare value is usually not enough.

Claim	Counter-example
Every prime is odd	2 is prime and even
Every odd number is prime	$9 = 3 \times 3$
$2^n - 1$ is prime for all $n \in \mathbb{N}$	$n = 4$: $2^4 - 1 = 15 = 3 \times 5$
$n^2 - n + 41$ is prime for all n	$n = 41$: the value is 41^2
If n is prime then $2^n - 1$ is prime	$n = 11$: $2^{11} - 1 = 2047 = 23 \times 89$
$x^2 > x$ for all real x	$x = \frac{1}{2}$: $\frac{1}{4} < \frac{1}{2}$

Why $n^2 - n + 41$ is the classic trap. It really does give primes for $n = 1, 2, \ldots, 40$ — testing "a few values" convinces you it is true. At $n = 41$,

$$41^2 - 41 + 41 = 41^2$$

which is divisible by 41. This is the standard warning that pattern-spotting is not proof.

5. Proof by contradiction

Proof by contradiction assumes the statement to be proved is *false*, and derives from that assumption something impossible. Since the assumption leads to an impossibility, it must have been wrong — so the original statement is true.

MEI names two proofs explicitly, and expects you to apply the method to unfamiliar statements as well.

5.1 The irrationality of $\sqrt{2}$

Theorem. $\sqrt{2}$ is irrational.
Proof. Suppose, for contradiction, that $\sqrt{2}$ is rational. Then we may write

$$\sqrt{2} = \frac{p}{q}$$

where p and q are integers, $q \neq 0$, and the fraction is **in its lowest terms** — that is, p and q share no common factor.

Squaring and rearranging:

$$2 = \frac{p^2}{q^2} \quad \Longrightarrow \quad p^2 = 2q^2$$

So p^2 is even. Now if p were odd, p^2 would be odd; hence p is even. Write $p = 2k$:

$$(2k)^2 = 2q^2 \quad \Longrightarrow \quad 4k^2 = 2q^2 \quad \Longrightarrow \quad q^2 = 2k^2$$

So q^2 is even, and by the same argument q is even.

But now p and q are *both* even, so they share the common factor 2 — contradicting our assumption that p/q was in lowest terms.

The assumption must therefore be false, so $\sqrt{2}$ is irrational. ■

The phrase **"in its lowest terms"** is not decoration — it is the assumption the contradiction violates. Omit it and there is no contradiction at all, and the proof scores very little. Equally, the step "p^2 even $\Rightarrow p$ even" needs its one-line justification (an odd number squared is odd); it is not self-evident.

Why the same argument fails for $\sqrt{4}$. Copy the proof with 4 in place of 2 and you reach $p^2 = 4q^2$. The step you would need is "p^2 divisible by 4 $\Rightarrow p$ divisible by 4" — and that is *false*: take $p = 2$, where $p^2 = 4$ is divisible by 4 but p is not. The proof breaks exactly where it should, since $\sqrt{4} = 2$ is rational. Understanding this is the difference between reciting the proof and knowing it.

The same structure proves $\sqrt{3}$ irrational, using the fact that $3 \mid a^2 \Rightarrow 3 \mid a$.

5.2 The infinity of primes

Theorem (Euclid). There are infinitely many prime numbers.
Proof. Suppose, for contradiction, that there are finitely many primes, and list them all:

$$p_1, \; p_2, \; p_3, \; \ldots, \; p_n$$

Consider the number

$$N = p_1 p_2 p_3 \cdots p_n + 1$$

Divide N by any prime p_i in the list. The product $p_1 p_2 \cdots p_n$ is divisible by p_i, so N leaves remainder 1. Hence **no prime in the list divides N**.

Now $N > 1$, so either N is itself prime, or it has a prime factor. In either case there is a prime not in the list — contradicting the assumption that the list contained every prime.

Therefore there are infinitely many primes. ■

A very common error is to claim "N must be prime". It need not be. With the list $2, 3, 5, 7, 11, 13$ we get $N = 30031 = 59 \times 509$ — composite, but both factors are primes *outside* the list, which is all the proof requires. Write "either N is prime or it has a prime factor not in the list".

5.3 Applying contradiction to unfamiliar statements

MEI states that contradiction should be applied to unfamiliar proofs, so expect statements you have not seen. The structure is always the same:

1. Write "Assume, for contradiction, that ..." and state the *negation* of what you want to prove.
2. Reason forwards, correctly, from that assumption.
3. Arrive at something impossible — a contradiction of the assumption itself, of a known fact, or of a definition.
4. Conclude: "the assumption is false, so the original statement is true".

Worked example. Prove that there is no smallest positive rational number.
Assume, for contradiction, that there is one; call it $r > 0$.

Consider $r/2$. Since r is rational, so is $r/2$; and since $r > 0$, we have $0 < r/2 < r$.

So $r/2$ is a positive rational smaller than r, contradicting the assumption that r was the smallest.

Hence no smallest positive rational number exists. ■

6. Negating a statement correctly

Since contradiction begins by assuming the negation, forming the negation accurately matters.

Statement	Negation
All swans are white	There exists a swan that is not white
$\forall x,\ P(x)$	$\exists x$ such that $P(x)$ is false
$\exists x$ such that $P(x)$	$\forall x,\ P(x)$ is false
$x > 3$	$x \leq 3$
There are finitely many primes	There are infinitely many primes

The negation of "all swans are white" is *not* "no swans are white". Over-negating is a frequent slip and destroys the argument, because you end up proving something stronger than you need — and usually something false.

7. Choosing a method

Situation	Method
A general algebraic identity or divisibility result	Deduction
The cases reduce to a small finite list (parity, remainders mod k)	Exhaustion
You are asked to show a statement is *false*	Counter-example
Irrationality, non-existence, "infinitely many", "no largest"	Contradiction
The direct route is awkward but the reverse implication is clean	Contrapositive

Reading the command word. "Prove" and "Show that" demand a complete general argument. "Verify" asks only that you check a particular case. "Disprove" asks for a counter-example. Answering a "prove" with a verification is one of the most costly misreadings in this topic.

8. Common pitfalls

Errors that repeatedly cost marks.

- Testing several values and concluding the general statement is true. That is not proof — see $n^2 - n + 41$.
- Using the same letter for two quantities that should be independent: writing two even numbers as $2n$ and $2n$ proves only the case where they are equal.
- Omitting "in lowest terms" in the $\sqrt{2}$ proof, leaving no contradiction to reach.
- Claiming $N = p_1 \cdots p_n + 1$ must be prime in Euclid's proof.
- Confusing a statement with its converse, or writing $\Leftrightarrow$ when only $\Rightarrow$ has been shown.
- Forgetting $q \neq 0$ when writing a rational as $\frac{p}{q}$.

- Failing to state that the cases in an exhaustion proof cover every possibility.
- Ending on algebra with no concluding sentence in words.